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Base-enabled access to diastereoselective spirofuran oxindoles and γ-functionalized allenoates†

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Base assisted divergent reactivity of isatins and allenoates has been achieved, which afforded diastereoselective spirofuran oxindoles and γ-functionalized allenoates. The DBU mediated Morita–Baylis–Hillman (MBH) reaction followed by the cascade annulation through the stabilized β-ammonium enolate intermediate led to the spiro-framework, wherein DABCO furnished the γ-functionalized allenoates. The protocol offers access to biologically relevant functionalized oxindole scaffolds with an excellent substrate scope under mild reaction conditions.

Oxindole is a vital pharmacophore present in many bioactive natural products and has attracted greater attention due to its significant pharmaceutical properties.[1] The functionalized and spirooxindole hybrids have also displayed various medicinal properties.[2] A few structures with a spirooxindole core are highlighted in Fig. 1. Spirotryprostatin A is an inhibitor of G2/M progression of cell division in mammalian cells.[3] The first macroline-related oxindole alkaloid Alstonisine is known to exhibit antimalarial properties,[4a] and several spirooxindole cores presently are being explored for antiviral therapeutic potential.[4b] Natural product inspired thiazolidinone spiro oxindole is a selective inhibitor of the *Mycobacterium tuberculosis* protein tyrosine phosphatase B.[5] On the other hand, a number of allenic motif containing compounds are also of pharmaceutical relevance (Fig. 1). For instance, allenic nucleoside analogs, adenallene, and cytallene are cytotoxic and antiviral agents, respectively. Hydroxyl acids and the methyl esters of allenes are known for their antifungal activity, and Grasshopper ketone is an allenic naturally occurring carotenoid[6] (Fig. 1).

Moreover, the construction of these hybrids is a challenge for synthetic organic chemists. Subsequently, many research groups have demonstrated the synthesis of oxindole based spiro cycles.[7] To highlight a few relevant reports, Jia *et al.* reported the synthesis of spirocyclic oxindole-butenolides by three component [2 + 2 + 1] cycloaddition of isocyanides, allenoates, and isatins (Scheme 1).[8] Lu *et al.*'s synthesis of chiral spirooxindoles was reported *via* an asymmetric [3+2] annulation reaction between allenic esters and isatinimines.[9] Shi and co-workers have carried out pioneering work in the synthesis of several privileged spirooxindole motifs.[10] Spiro tetrahydrofuran oxindoles by the reaction of allenoate phosphine zwitterions with isatins were developed by Nair's – group in 2015 (Scheme 1).[11] A three-component reaction by cyclization of isocyanide, allenoates, and isatins has led to the structurally complex spirooxindoles.[12] The literature reports emphasize that the majority of the spirooxindole chemistry is associated with the spiro pyrrole/pyrrolidine and spiro pyridine oxindoles.[7,9,10] Hence, spirofuran oxindoles are relatively less explored for their biological properties; this signifies the need of novel methodologies for their synthesis.

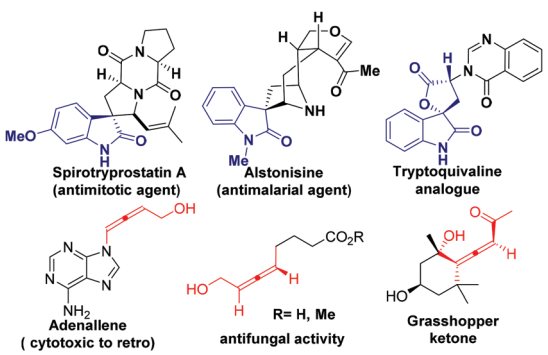


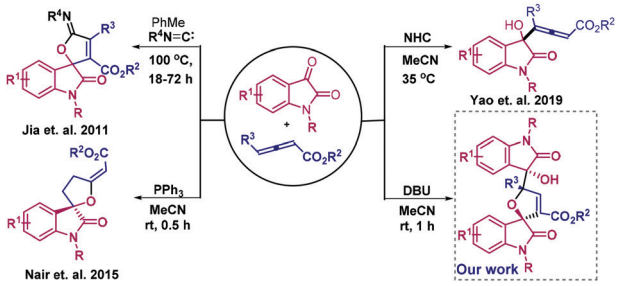
Fig. 1 Selected bioactive oxindole and allene scaffolds.

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† Electronic supplementary information (ESI) available: Experimental details, compound characterization, and NMR spectra. CCDC 1979517 (**3ag**) and 1979628 (**4h**). For ESI and crystallographic data in CIF or other electronic format see DOI: 10.1039/d0cc07715a

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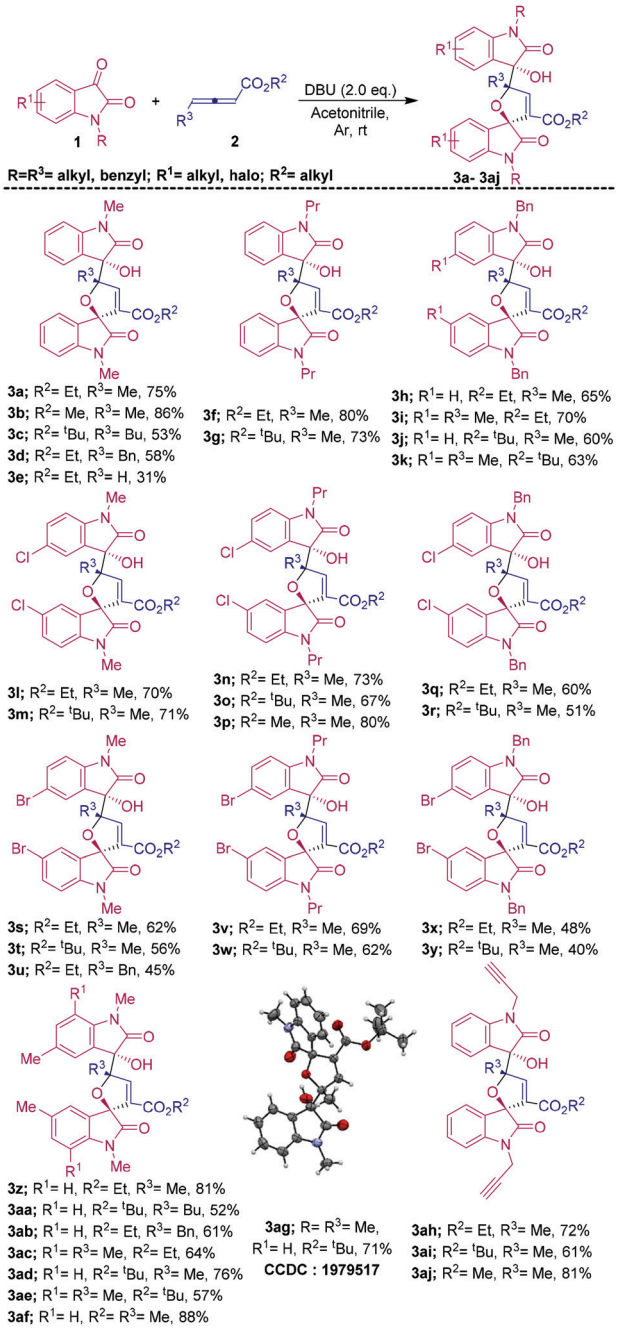


Scheme 1 Reactivity of isatins with allenoates.

In view of these exciting observations and in continuation of our attention towards the spiro and biologically relevant heterocycles,[13] it was of great interest to investigate isatin reactivity towards allenoate zwitterions with reasonable expectations of accessing spirofuran oxindoles in the process. At the outset, the protocol's feasibility was first investigated by the reaction of *N*-methyl isatin **1** and 3-methyl allenoate **2** in the presence of 1,8-Diazabicyclo [5.4.0]undec-7-ene (DBU, 1.0 equiv.) in acetonitrile at room temperature. After 2 h, the spirofuran oxindole **3a** was obtained in 18% yield as a single diastereomer. Its molecular structure and relative stereochemistry were established by Nuclear Magnetic Resonance (NMR), and High-Resolution Mass (HRM) spectroscopy, and further confirmed *via* single-crystal X-ray analysis of one of the derivatives **3ag** (Scheme 2).

The reaction parameters were screened to improve the yield and overall process; the results are summarized in Table 1. Firstly, we tuned the stoichiometric ratio of substrates; 1 equiv. of *N*-methyl isatin **1** and 0.6 equiv. of 3-methyl allenoate **2** resulted in a 43% yield of **3a**. An excess of allenoate (Table 1, entry 1–4) suppressed the yield. Increment in the DBU loading to 2 equiv. afforded 56% yield (Table 1, entry 8). Furthermore, an increase or decrease in DBU loading from 2 equiv. and catalytic amounts (Table S2, ESI†) did not show any positive results. Subsequent screening of bases like DMAP, Et_3N, and DABCO demonstrated that DBU is the best choice (Table 1, entry 8). Careful screening of the solvents and neat reaction (Table 1, entries 15–21) indicated that acetonitrile is the most suitable solvent. Further assessment on variation in the reaction temperature (Table 1, entries 22 and 23) lowered the yields. Gratifyingly, the time-dependent quenching of the reaction between 0.5 h and 2 h markedly resulted in an improved yield of 75% (Table 1, entry 25). By extending the reaction time, the yield decreased to some extent (Table S2, ESI,† entry 5); this is due to the formation of complex products (observed in TLC). Based on these experimental results, the best reaction condition was confirmed to be 1 equiv. of *N*-methyl isatin, 0.6 equiv. of allenoate and 2 equiv. of DBU in acetonitrile at room temperature for 1 h (Table 1, entry 25).

With the optimized conditions in hand, the reaction's versatility has been successfully explored for diversely substituted isatins with various allenoates (Scheme 2). In all the experiments, the reaction proceeded smoothly and furnished the expected products in moderate to good yields. However, the highest yield

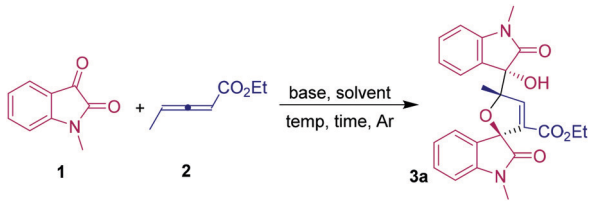


Scheme 2 Scope of the reaction for spirofuran oxindoles.

of 88% was noticed for compound **3af.** Slightly lower yields were observed in the case of *N*-protected isatins with a bulkier benzyl group and also for *N*-protected 5-Bromo isatins. Moreover, it was found that allenoates having smaller ester substituents such as Me and Et, attributed comparatively higher yield than the bulkier ^{t}Bu group. Comparably moderate to good yields were obtained for variously γ-substituted allenoates. 5-OMe/5-NO_2 isatins and γ-Phenyl allenoates did not furnish the expected cyclized product. These results highlight the broad substrate scope of DBU-base mediated MBH type reaction followed by cascade annulation of isatins and allenoates.

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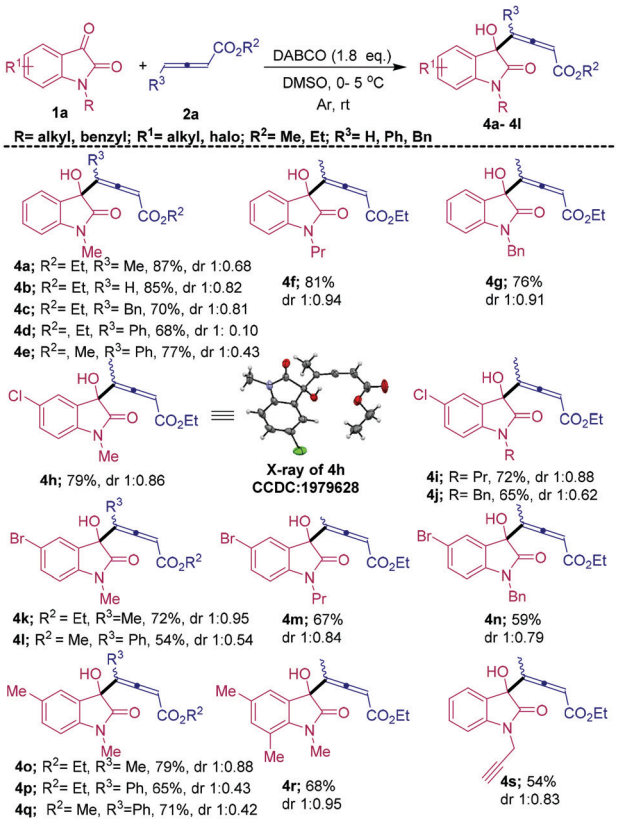
Table 1 Optimization of the reaction[a]



Entry	2 (equiv.)	Base (equiv.)	Solvent	Temp. (°C)	Time (h)	Yield (%)[c]
1	1	DBU (1.0)	Acetonitrile	r.t.	2	18
2	1.5	DBU (1.0)	Acetonitrile	r.t.	2	27
3	2.0	DBU (1.0)	Acetonitrile	r.t.	2	10
4	3.0	DBU (1.0)	Acetonitrile	r.t.	2	Trace
5	0.6	DBU (1.0)	Acetonitrile	r.t.	2	43
6	0.5	DBU (1.0)	Acetonitrile	r.t.	2	32
7	0.6	DBU (1.5)	Acetonitrile	r.t.	2	49
8	0.6	DBU (2.0)	Acetonitrile	r.t.	2	56
9	0.6	DBU (2.5)	Acetonitrile	r.t.	2	28
10	0.6	DBU (0.5)	Acetonitrile	r.t.	2	Trace
11	0.6	DMAP (2.0)	Acetonitrile	r.t.	24	—
12[b]	0.6	DABCO (2.0)	Acetonitrile	r.t.	24	—
13	0.6	Et_3N (2.0)	Acetonitrile	r.t.	24	—
14	0.6	PPh_3 (2.0)	Acetonitrile	r.t.	24	—
15	0.6	DBU (2.0)	Toluene	r.t.	2	Trace
16	0.6	DBU (2.0)	1,4-Dioxane	r.t.	2	Trace
17	0.6	DBU (2.0)	DCM	r.t.	2	15
18	0.6	DBU (2.0)	THF	r.t.	2	7
19	0.6	DBU (2.0)	EtOH	r.t.	2	—
20	0.6	DBU (2.0)	Benzonitrile	r.t.	2	22
21	0.6	DBU (2.0)	Neat	r.t.	2	Trace
22	0.6	DBU (2.0)	Acetonitrile	0–5	2	24
23	0.6	DBU (2.0)	Acetonitrile	50	2	31
24	0.6	DBU (2.0)	Acetonitrile	r.t.	0.5	33
25	**0.6**	**DBU (2.0)**	**Acetonitrile**	**r.t.**	**1**	**75**
26	0.6	DBU (2.0)	Acetonitrile	r.t.	1.5	66

[a] All the reactions were conducted with 1 equiv. of **1** (0.31 mmol). [b] DABCO afforded product **4a** in 36% yield. [c] Isolated yield.

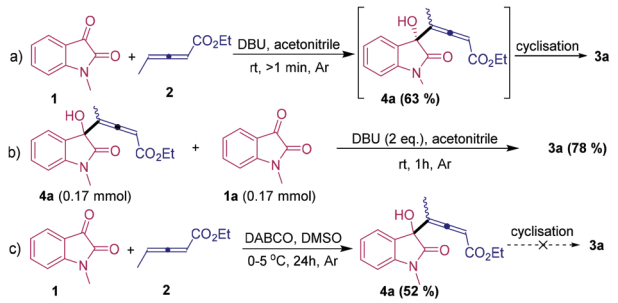
During the optimization of spirofuran oxindole synthesis, a serendipitous observation led to the identification of γ-functionalized allenoate **4a** (Scheme 3), and it was found that the reaction between *N*-methyl isatin **1** and allenoate **2** in the presence of (1,4-diazabicyclo[2.2.2]octane (DABCO)) in acetonitrile could not furnish the spirofuran oxindoles. Nevertheless, a different product with a slight change in retention factor (R_f) value (observed in TLC) was separated and characterized as γ-functionalized allenoate (**4a**). The structure was unambiguously confirmed through the single-crystal X-ray analysis of one of the analogs **4h** (Scheme 3). Allenes and their functionalized hybrids were found to be explored for various therapeutic applications.[6] Very recently, Yao *et al.* reported an NHC-catalyzed regiospecific γ-functionalized allenoate (Scheme 1).[14] Encouraged by the above results, a systematic optimization has been carried out by considering the stoichiometric ratio of substrates, catalyst loading, solvent suitability, *etc.*, to improve the yields (ESI,† Table S1). After careful observation, 1 equiv. of *N*-methyl isatin **1**, 1.5 equiv. of allenoate **2** and 1.8 equiv. of DABCO in DMSO at 0–5 °C for 0.5 h (ESI,† Table S1, entry 21) is considered as the best-optimized condition. Furthermore, we explored the scope of the reaction for various γ-functionalized allenoates under the optimized conditions.



Scheme 3 The generality of the reaction for γ-functionalized allenoate oxindoles.

Including bulky groups such as propyl and benzyl groups on the nitrogen atom of isatin could furnish the expected product in 59–87% yields, and propargylated isatin gave the product **4s** in 54% yield (Scheme 3).

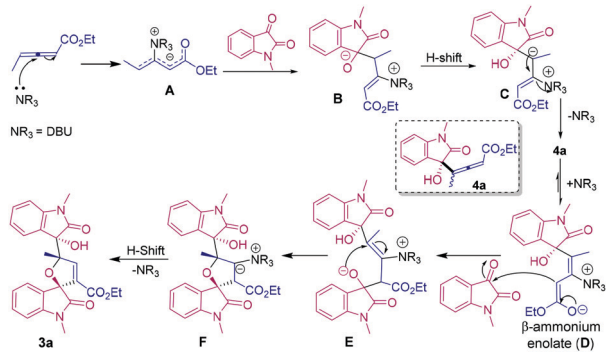
To gain preliminary insight into the reaction mechanism and resolve how two different bases resulted in a different reactivity, we performed a few control experiments (Scheme 4 and Table S2, entries 6, 10, ESI†). In a DBU mediated reaction, within a minute, we have observed the formation of product **4a** (Scheme 4a); and this was followed by cascade cyclization between quickly formed **4a** and one more molecule of isatin **1a** in the presence of the stoichiometric amount of DBU leads to the formation of **3a**. To achieve faster rate, a higher concentration of enolate is required. Here, the DBU will shift the



Scheme 4 Control experiments.

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Scheme 5 Plausible mechanistic pathway.

equilibrium towards the β-ammonium enolate (Scheme 5) by stabilizing it through conjugation, which increases its equilibrium concentration; eventually, the reaction rate is enhanced for further cyclisation.[15] The DBU mediated reaction between isolated **4a** and **1a** further confirms the involvement of **4a** in the formation of **3a** (Scheme 4b). As shown in Scheme 4c, the lack of conjugation in DABCO, substantial reduction in nucleophilicity by incorporating bulky substituents α to nitrogen results in steric hindrance to the β-ammonium enolate.[15] This causes a shift in the equilibrium back to **4a**. Therefore it could not initiate further cyclization even after 24 h (Table S2, entry 10, ESI†).

Based on the experimental results and considering the previous reports,[11,14,15] we have proposed a plausible mechanism for spirofuran oxindole formation. Initially, the nucleophilic addition of DBU to 3-alkyl allenoate **2** occurs, affording a zwitterion intermediate **A**. The zwitterion **A** thus formed can easily add to the carbonyl carbon of the *N*-protected isatin **1** to form the intermediate **B**. After that, the proton transfer delivers **C**; the elimination of DBU results in an intermediate product **4a**, which eventually forms stabilized β-ammonium enolate intermediate **D** in the presence of a stoichiometric amount of base. The latter undergoes nucleophilic addition to the second molecule of isatin furnishing **E**, which upon cyclization, leads to **F**. This intermediate, after proton transfer and elimination of DBU, delivers the desired product **3a** (Scheme 5). Similarly, the plausible mechanism for γ-functionalized allenoate (**4**) is presented in Scheme S2 (ESI†).

In summary, base mediated divergent reactivity of isatins and allenoates has been accomplished. These protocols offer a facile synthetic route to access the spiro and functionalized oxindoles. In particular, DBU promoted MBH type reaction provides unprecedented access to the spiro-furan oxindoles of medicinal importance. The method proceeds under mild reaction conditions and complements a diverse substrate scope.

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Conflicts of interest

There are no conflicts to declare.

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